E171: Dynamics of Elastic Systems Final Project

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1 Introduction

1.1 System setup

We are given a model of an one-dimensional nominally periodic structure. The goal of this project is to simulate a bladed disk where each oscillator is representative of a singular blade. This system consists of un-damped spring mass systems with equivalent stiffness $k_i s$.. In terms of the bladed disk, stiffness $k_i s$ represents the flexural rigidities of the blade. coupled with stiffness k_c. The model of the system is shown below in Figure 1.

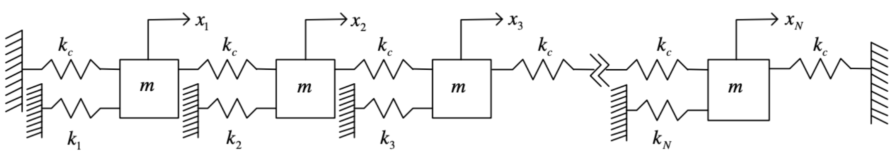

Figure 1: NDOF System Model

By analyzing this bladed disk via the model, we hope to gain insight on the movement of rigid bodies as well as better understand NDOF systems.

2 Perfectly Periodic System

First, we will examine the dynamics of an ordered periodic chain to eventually compare the results of a perfectly periodic system with those of a disordered system. Let's assume that $k_1 = k_2 = \ldots = k_N = k_0 =$ constant.

2.1 Equations of Motion

We can determined the equations of motion by first analyzing a two degree of freedom (2DOF) spring mass model system. The system model is given in Figure 2. below.

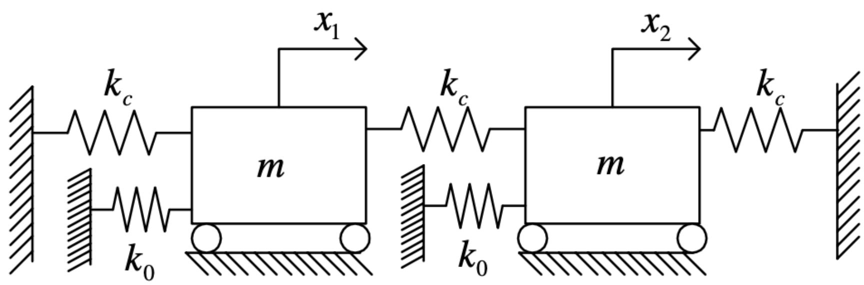

Figure 2: 2DOF System Model

We use the Newtonian equations of motion method to achieve the following system of equations in matrix form.

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2k_c + k_0 & -k_c \\ -k_c & 2k_c + k_0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{1}$$

We then extend this 2DOF system of equations to an NDOF system by accounting for the possibility of N number of rows and columns (corresponding to N oscillators). This is accounted for in the system of equations in matrix form below.

$$\begin{bmatrix} m & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & m \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \vdots \\ x_N \end{bmatrix} + \begin{bmatrix} 2k_c + k_0 & -k_c & & \mathbf{0} \\ -k_c & \ddots & \ddots & \\ & \ddots & \ddots & -k_c \\ \mathbf{0} & & -k_c & 2k_c + k_0 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \tag{2}$$

Note that each oscillator is attached to two coupling springs (k_c) and one flexural stiffness (k_0, like the 2DOF system). Each coupling spring either attaches to a wall or to a neighboring oscillator, so there can be up to two coupling terms for each oscillator, one for each neighbor. These requirements result in [K] being a tridiagonal matrix.

2.2 Nondimensionalize

The next step is to nondimensionalize the equations of motion by diving through by k_0. We do so by introducing the dimensionless natural frequencies given by,

$$\bar{\omega}_r = \omega_r \sqrt{m/k_0}, r = 1, \ldots, N \tag{3}$$

Where ω_r is the rth natural frequency. Furthermore, we introduce a dimensionless coupling ratio $R = k_c/k_0$. Substituting R into equation (2) we get

$$\begin{bmatrix} \frac{m}{k_0} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \frac{m}{k_0} \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \vdots \\ x_N \end{bmatrix} + \begin{bmatrix} 2R+1 & -R & & \mathbf{0} \\ -R & \ddots & \ddots & \\ & \ddots & \ddots & -R \\ \mathbf{0} & & -R & 2R+1 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{4}$$

This equation now allows us to solve this Generalized Eigenvalue Problem:

$$\omega_i^2 \left[M\right] U_i = \left[K\right] U_i. \tag{5}$$

Matrix M is equal to m * matrix I. We substitute this in to get:

$$\omega_i^2 m \left[I\right] U_i = \left[K\right] U_i. \tag{6}$$

Divide the equation by k_0:

$$\omega_i^2 \frac{m}{k_0} \left[I\right] U_i = \frac{1}{k_0} \left[K\right] U_i. \tag{7}$$

Now, using equation (3) to rewrite equation (7), we get:

$$\bar{\omega}_i^2 U_i = \frac{1}{k_0} \left[K\right] U_i. \tag{8}$$

We can now solve for the mode shapes with ease using MATLAB with the resulting eigenvalue problem.

2.3 Modes of Vibration for a 50 Degree-of-Freedom System

We used a Matlab script to solve for the mode of vibration for a chain of 50 degrees of freedom system. The graph for the first 5 mode shapes and when r = 0.01 is given below in Figure 3.

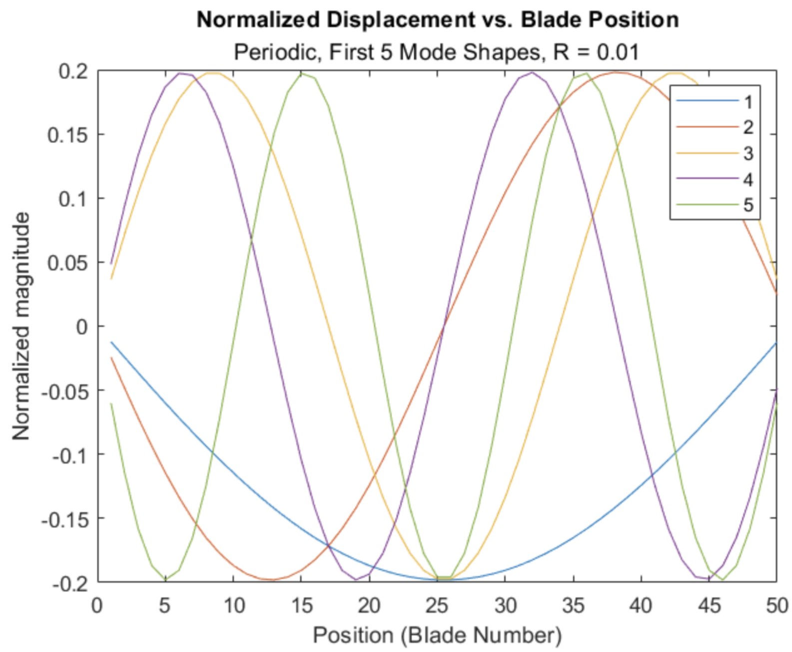

Figure 3: Mode shapes of perfectly periodic system.

2.4 Analysis Concerning the Displacements of the Perfectly Periodic Chain

Looking at figure 3 we can observe a patterned, regular behavior for the mode shapes. Each mode shape is a sinusoidal wave, with each increasing mode of vibration having a corresponding mode shape with one more standing node. Like the ruler demo performed earlier in class, and like any continuous medium upon which a standing wave can occur, more standing nodes correspond to a higher-energy vibration. This perfectly periodic setup allows for perfect standing wave behavior.

2.5 Analysis of Changing the Coupling Ratio R

We plot various graphs when R equals different values. We chose to look at when R = 0.01, R = 0.1, and R =1. The plots are given side by side below in Figure 4.

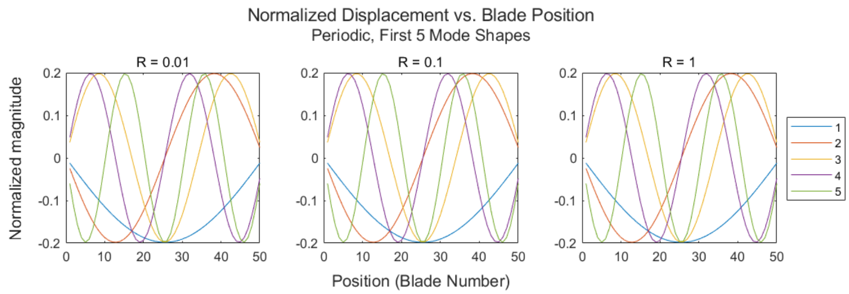

Figure 4: The relationship between normalized displacement and blade position when R = 1, R = 0.1 and R = 0.01

We can also look at the relationship between the normalized frequency and the mode of vibration number. The graphs of this are plotted below in Figure 5.

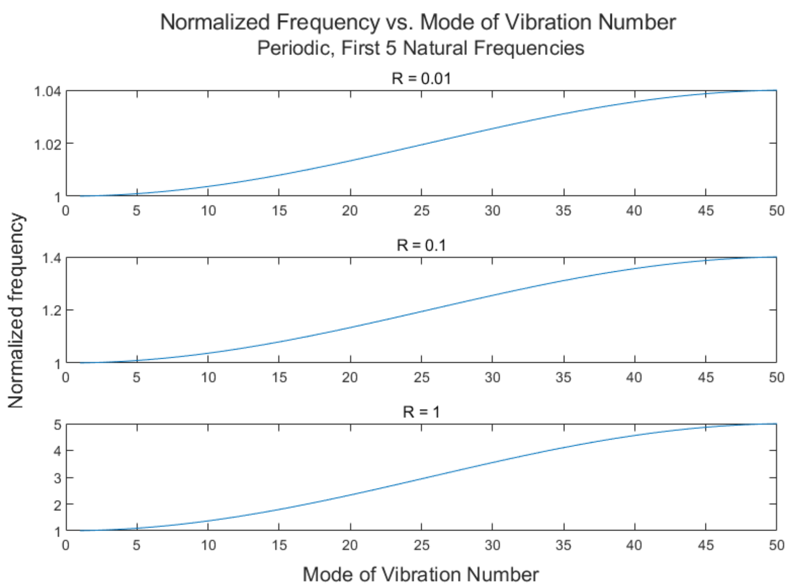

Figure 5: The Normalized Frequency and Mode Vibration Number Relationships when R = 0.01, R = 0.1, and R = 1

There is no change in the relationship between normalized displacement and blade position as R is adjusted. This is because the we would expect that the modes would not change shape, rather the magnitude does. However, R does change the range of natural frequencies, as increasing the highest frequency led to an increase in R. This means that there is a relationship between stiffness increase and a higher energy mode shape: As the stiffness of coupling increases compared to the stiffness of each blade, the higher energy mode shapes are at greater frequencies.

3 Nearly Periodic System

Next, we will analyze the dynamical characteristics of a nearly periodic system. Let the $k_i s$ be defined by

$$k_i = k_0(1 + \epsilon\delta k_i), i = 1, ..., N \tag{9}$$

where k_0 denotes the nominal stiffness, ϵ represents the disorder strength, and the δ k_is corresponds to the disorder parameter of the ith component site. Since the irregularities are not known exactly, the k_i's are treated as random variables. Furthermore, if the δ k_i's are random variables with zero mean and variance of one, the absolute value of the parameter ϵ becomes an estimate of the standard deviation of the irregularities.

3.1 Find the Eigenvalue Problem for the Disordered System

We will now reformulate the equations of motion to incorporate the delta term for each of the stiffness. With our arbitrary NDOF matrix-form system of equations:

$$\begin{bmatrix} m & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & m \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \vdots \\ x_N \end{bmatrix} + \begin{bmatrix} 2k_c + k_1 & -k_c & & \mathbf{0} \\ -k_c & \ddots & \ddots & \\ & \ddots & \ddots & -k_c \\ \mathbf{0} & & -k_c & 2k_c + k_N \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{10}$$

Now insert $k_i = k_0(1 + \epsilon\delta k_i), i = 1, \ldots, N$ into the K matrix:

$$\begin{bmatrix} m & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & m \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \vdots \\ x_N \end{bmatrix} + \begin{bmatrix} 2k_c + k_0(1+\epsilon\delta k_1) & -k_c & & \mathbf{0} \\ -k_c & \ddots & \ddots & \\ & \ddots & \ddots & -k_c \\ \mathbf{0} & & -k_c & 2k_c + k_0(1+\epsilon\delta k_N) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{11}$$

Nondimensionalizing with R and $\bar{\omega}_r$ as before, we get:

$$\begin{bmatrix} \frac{m}{k_0} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \frac{m}{k_0} \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \vdots \\ x_N \end{bmatrix} + \begin{bmatrix} 2R + 1 + \epsilon\delta k_1 & -R & & \mathbf{0} \\ -R & \ddots & \ddots & \\ & \ddots & \ddots & -R \\ \mathbf{0} & & -R & 2R + 1 + \epsilon\delta k_N \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{12}$$

This gives us the eigenvalue problem below:

$$\bar{\omega}_i^2 U_i = ([K_0] + [\delta K])U_i. \tag{13}$$

3.2 Mistuned Mode Shapes

The mode shapes in a nearly periodic system are said to be mistuned. Using the formulation provided, we developed a set of mistuned parameters using δk_i's.

We plot the comparison between mistuned mode shapes and perfect mode shapes below.

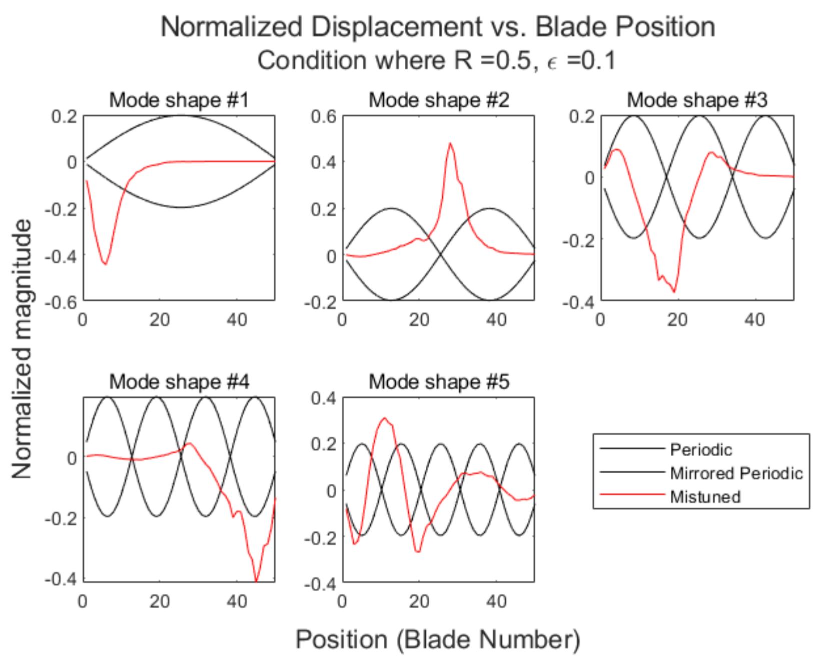

Figure 6: R = 0.5, ϵ = 0.1 to Compare Mistuned Mode Shapes with Perfect Mode Shapes

The comparison above reveals a drastic change in mode shape, with the periodic standing waves replaced with spiked shapes, centering on a few blades. The early mode shapes are especially aperiodic, with their behavior characterized almost entirely with a spike around a few blades. As the mode shape number increases the spikes are less intense, but they still exhibit spike behavior.

3.3 Localization Emerges

Localization behavior can be studied across the different mode shapes. The emergence of the localization is plotted in Figure 7 when R = 0.01 and ϵ = 0.1.

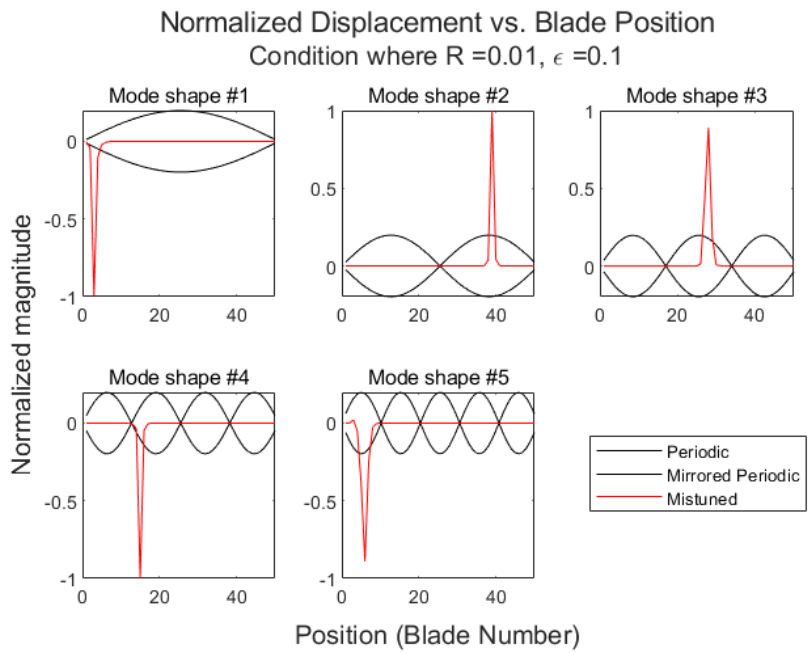

Figure 7: R = 0.01, ϵ = 0.1 to Compare Mistuned Mode Shapes with Perfect Mode Shapes

Note that each these mode shapes are purely a single spike, as if each mode shape is only a single blade selected to vibrate.

3.4 Perturbation for Physical Insight

To perform perturbation, we use the equations derived in class to calculate the $[\delta U]$ and $[\delta\lambda]$ matrices to calculate $[U]$ and $[\lambda]$. We found that, in some cases, splitting larger perturbation calculations into smaller steps and progressively calculating got a better result, so we added a *steps* parameter to control how many steps each calculation is performed in. To reveal localization, we chose to mistune a single blade, blade 20.

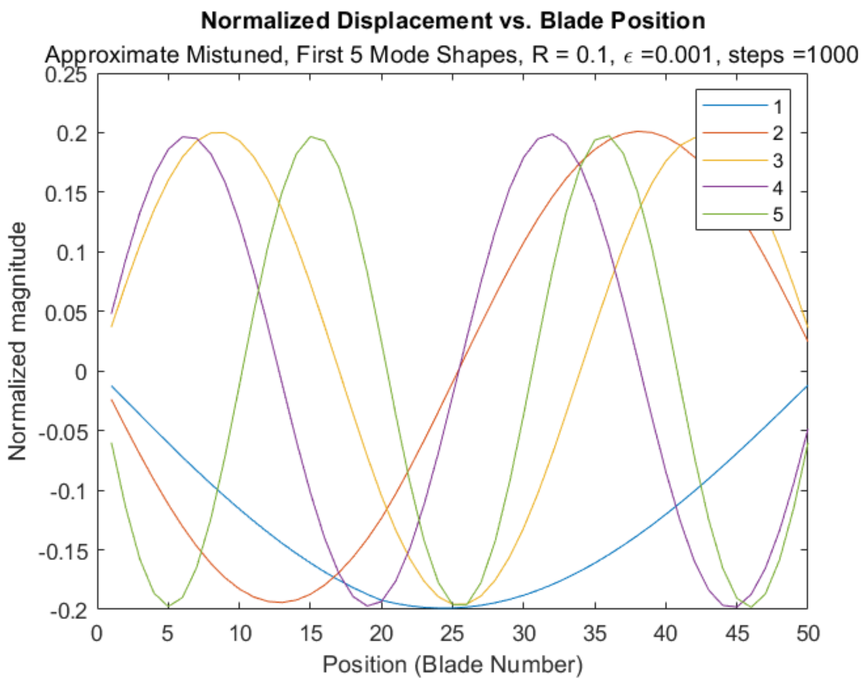

Figure 8: Effect of mistuning blade 20 with a small ϵ on mode shapes.

In the case where R is much larger than ϵ, we find that no noticeable localization occurs.

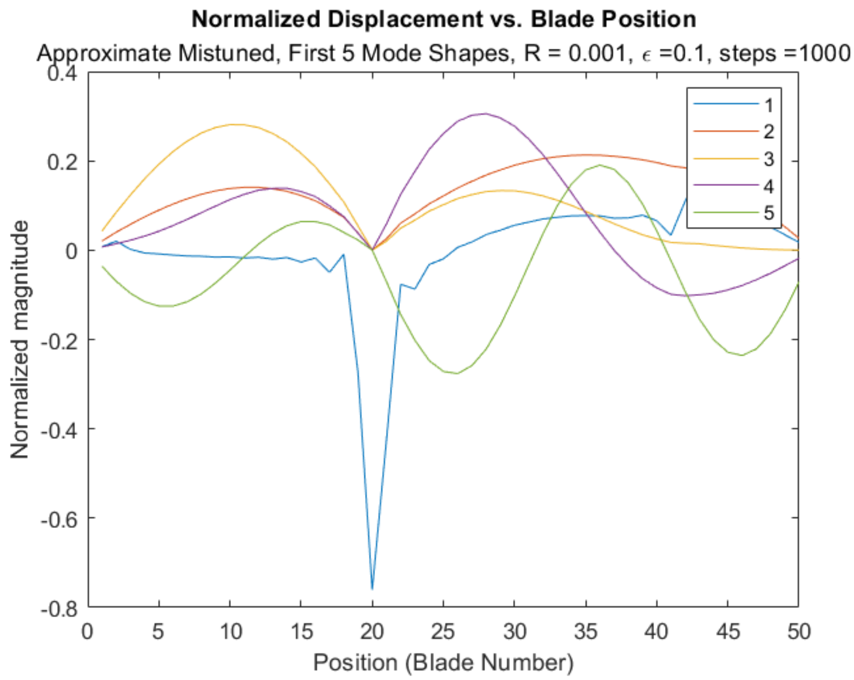

Figure 9: Effect of mistuning blade 20 with a small R on mode shapes.

When R is much smaller than ϵ, significant localization occurs. This localization is noticeable as the large spike at position 20.

3.5 Perturbation vs. Resolving

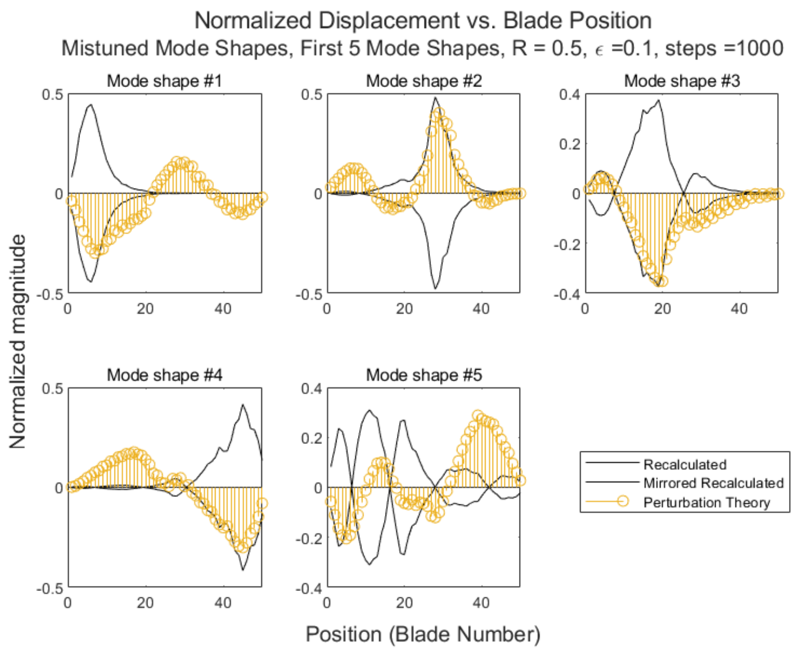

Figure 10: Replication of 3.2, comparing eigenvectors.

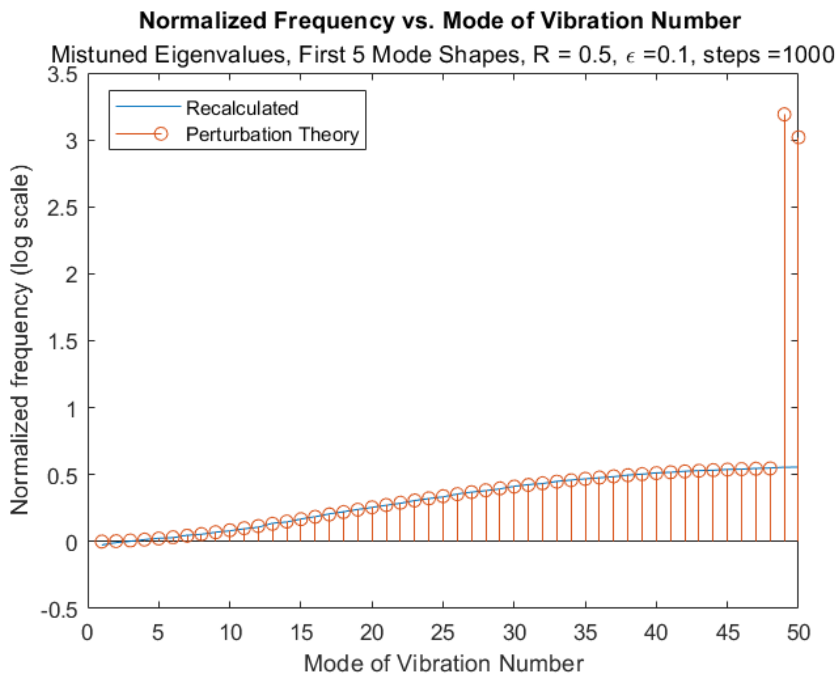

Figure 11: Replication of 3.2, comparing eigenvalues.

With R larger than ϵ we find that perturbation works well for approximating the changes. Note that the mode shapes match pretty well, and the frequencies are close as well.

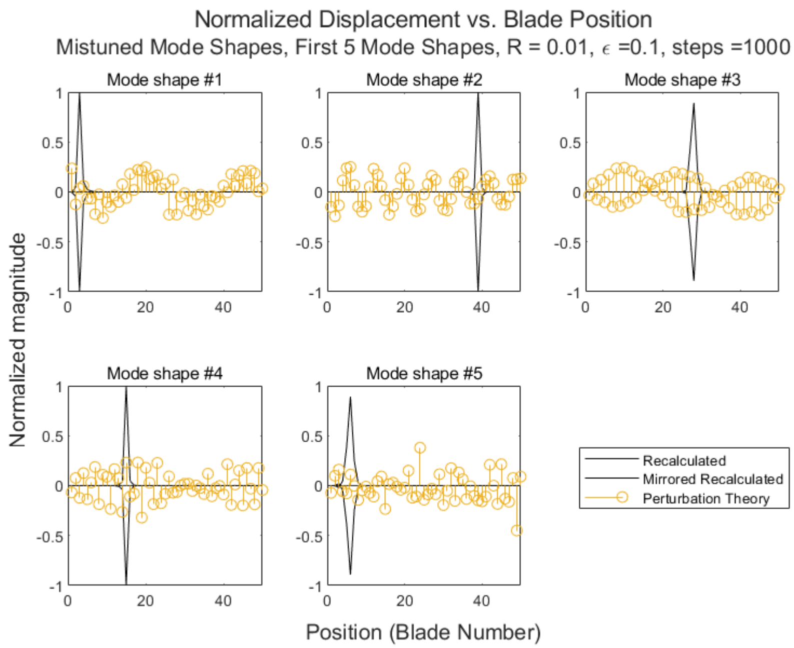

Figure 12: Replication of 3.3, comparing eigenvectors.

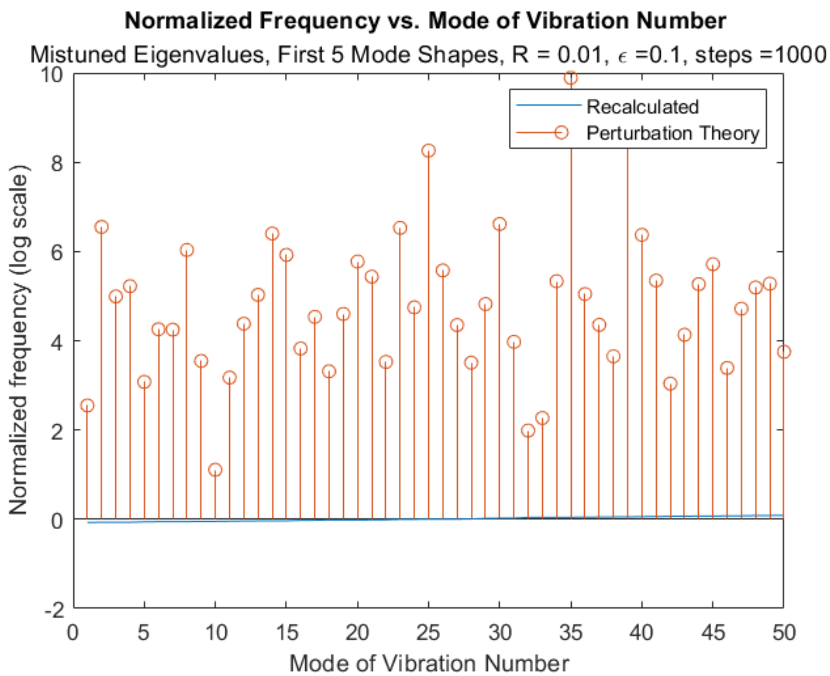

Figure 13: Replication of 3.3, comparing eigenvalues.

However, with a smaller R compared to ϵ we find that perturbation struggles to approximate this 50 DOF system. Note that neither the frequencies nor the mode shapes match well.

4 Conclusion

There are three main questions that we will answer in this part of the report. 1. Knowing that the total energy of each component is proportional to the its displacement squared, what is the implication of the findings above? 2. How can we harness the localization phenomena to our advantage? 3. How can we minimize the localization phenomenon?

When there is more displacement in the system, then there will be an even larger amount of energy to be dissipated. When this dissipation does not occur properly for large displacements, the blades will most likely break. We can harness the localization phenomena to our advantage by applying displacement to specific areas that we want to study in order to see the results more clearly. For instance, if there is a blade position that we wish to study, then increasing the mistuning will allow there to be a larger displacement, and thus that specific area can be studied. This study could lead to a blade being mistuned on purpose to be a designated energy dissipation site, resulting in less energy dissipation on other blades. We can minimize the localization phenomena by increasing the quality of the blades, resulting in a more consistent stiffness. This is evident in increasing the R value.

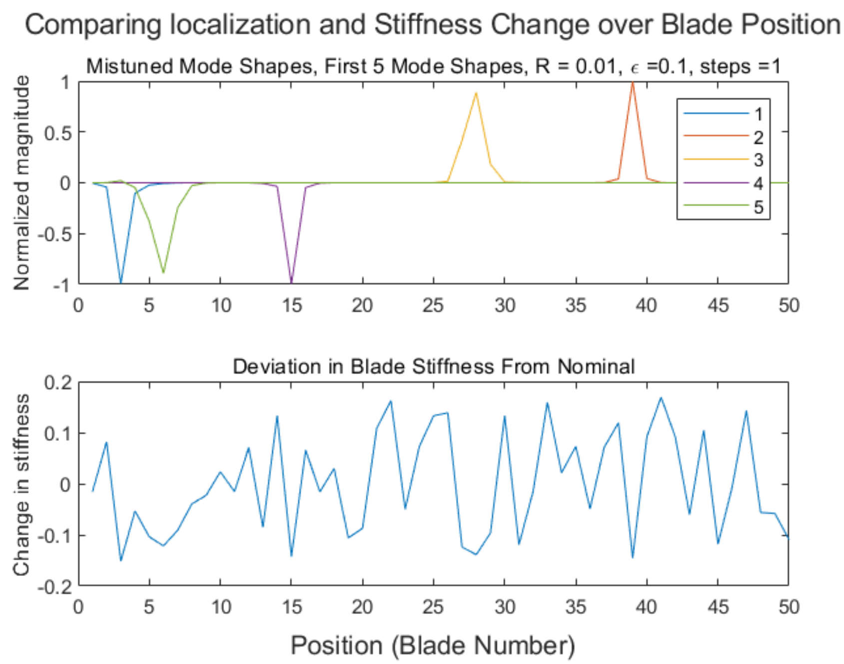

Figure 14: Relating Mode Shape to Mistuning

The above figure shows how mistuning can lead to mode localization, as the first 5 mode shapes correspond to spots where a specific blade is locally quite different in stiffness. Overall, we find that localization occurs when the periodic structure is disturbed, and that perturbation theory allows for study of a few mistunings to see the effect of changing one blade.